

# Antoine Boucher

SOFTWARE ENGINEER — PLATFORM AND COMPUTER GRAPHICS

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## Summary

Software engineer on platform engineering and computer graphics. Day-to-day work spans Kubernetes, GCP, Terraform, GitLab CI/CD, and observability for Java/Kotlin/Spring Boot and Python microservices, with JUnit, Playwright, and DevSecOps on releases. Master's at ÉTS: real-time rendering, physics simulation, and tactile sensing in C++, OpenGL/GLSL, and Blender.

## Competences

|                                |                                                                                 |
|--------------------------------|---------------------------------------------------------------------------------|
| <b>Languages</b>               | Java, Kotlin, Python, C++, JavaScript                                           |
| <b>Backend &amp; APIs</b>      | Spring Boot, microservices · Int.: C#, .NET, Android, GraphQL                   |
| <b>Graphics / 3D</b>           | OpenGL, GLSL, C++, real-time rendering · Int.: Blender, CUDA, OpenCV            |
| <b>Web</b>                     | Angular, React, TypeScript                                                      |
| <b>Cloud &amp; platform</b>    | Docker, Kubernetes, Terraform, GCP, Azure, AWS                                  |
| <b>Delivery</b>                | Git, GitLab CI/CD, GitHub Actions · Int.: Azure DevOps                          |
| <b>Data</b>                    | PostgreSQL, MongoDB, Redis, Kafka, SQL                                          |
| <b>Reliability</b>             | Playwright, JUnit, ELK, Grafana, Prometheus · Int.: Sentry                      |
| <b>Security &amp; practice</b> | Linux, OWASP, Agile/SCRUM/Kanban, uml-mcp maintainer, MCP                       |
| <b>Credentials</b>             | AWS CCP (to Apr 2027); QcES; CQ CP, Data, ML, Serverless, Networking, Architect |

## Experience

### IMC2 – Institut multidisciplinaire en cybersécurité et cyberrésilience

Montréal, QC (Remote)

Research Assistant — Secure cloud platforms (Permanent Part-time)

Mar 2025 – Feb 2026

- Split a monolith into per-service repos and ran 5 GitLab CI/CD pipelines plus runners on two GKE clusters in GCP (Terraform networking and observability) for repeatable releases
- Containerized workloads in Docker, ran research GKE on GCP (workload networking), and deployed services with Terraform so infra changes were easier to ship safely
- Added JUnit and Playwright to 5 Java microservices; promoted Android builds through GitLab into the Terraform-managed GKE stack so mobile and backend releases stayed aligned
- Built scripted risk analysis over 1000+ risks in 181+ threat categories (weighted scoring, star-based prioritization, Angular triage UI) so analysts could prioritize with consistent context
- Technical Skills:** Java, Kotlin, Docker, Kubernetes, Terraform, GitLab CI/CD, GCP, JUnit, Playwright, Angular
- Soft Skills:** Cross-functional collaboration, stakeholder communication, analytical prioritization

### Polytechnique Montréal

Montréal, QC (On-site)

Teaching Assistant — DevSecOps (LOG 8100) (Contract Part-time)

Sep 2024 – Jan 2026

- Built and maintained the GitLab workspace (repos, CI/CD templates, runners, workflows) for approximately 20 concurrent teams, cutting admin overhead during the term
- Coached Docker and Kubernetes build, deploy, and debug under LOG 8100; used OWASP-driven examples tied to hands-on lab scenarios
- Refreshed DevSecOps-aligned lab artifacts, graded practicals, and coached resubmissions for faster, cleaner turnaround
- Technical Skills:** GitLab, CI/CD, Docker, Kubernetes, OWASP, Linux
- Soft Skills:** Team coaching, constructive feedback, classroom facilitation

### École de technologie supérieure (ÉTS)

Montréal, QC (Hybrid)

Teaching Assistant (Contract Part-time)

Feb 2024 – Apr 2026

- Delivered labs, integrator-project support, and office hours on mobile/UX, integrator-project, and distributed-data tracks, totaling approximately 150 contact hours (e.g., TCH 057, IND 500, TCH 099, GTI 660, GTI 320)
- Authored and refreshed SQL, MongoDB, Kafka, and PostgreSQL exercises for IND 500, GTI 660, and related cohorts, with rubrics aligned to course outcomes
- Unblocked integrator teams on architecture and backend/data; kept Azure-backed course infrastructure running; SCRUM coaching on mobile/UX and GTI 320 so students could ship end-to-end features
- Technical Skills:** SQL, MongoDB, Kafka, PostgreSQL, Azure, Android Studio, SCRUM
- Soft Skills:** Student mentorship, rubric-driven feedback, patience under deadline pressure

## IONODES

Cloud Developer (Permanent Full-time)

Laval, QC (Hybrid)

May 2023 – Aug 2023

- Designed a three-tier Auth0 subscription model for organization-scoped IoT access, gating product capabilities per tenant
- Integrated Sentry in C# with Serilog for structured production errors, making production issues easier to trace for support and engineering
- Advanced ONVIF and WebRTC streaming at IoT scale; ran Azure DevOps with feature flags and repeatable promotions for safer releases
- Technical Skills:** C#, .NET, Azure DevOps, Auth0, Sentry, ONVIF, WebRTC
- Soft Skills:** Requirements clarification, partner communication, calm under production pressure

## Intact

Back End Developer (Co-op)

Montréal, QC (Hybrid)

Jan 2022 – May 2022

- Delivered a 6-step signup flow and mobile-app observability on Kotlin/Spring Boot microservices serving 10,000+ active users, improving activation for new accounts
- Implemented Kotlin login and MFA APIs for client mobile apps while preserving policy controls
- Extended the ELK stack with Kibana dashboards and harmonized microservice error codes so operations could triage incidents faster
- Technical Skills:** Kotlin, Spring Boot, GraphQL, MongoDB, ELK, Kibana
- Soft Skills:** Agile teamwork, incident communication, cross-team coordination

## Divine Essence (Union-Nature)

Web Specialist (Contract Part-time)

Dorval, QC (On-site)

Jun 2020 – Dec 2020

- Modernized the bilingual CommerceBuild storefront (FR/EN blog navigation, accessibility, SEO sales labels, custom Add-to-Cart API, 3CX chatbot), bringing more qualified traffic and self-serve support
- Technical Skills:** HTML, CSS, JavaScript, CommerceBuild, SEO
- Soft Skills:** Client communication, UX empathy, staff training

## Power Go

Full Stack Web Developer (Co-op)

Montréal, QC (Remote)

Sep 2020 – Apr 2021

- Built a Python/Django/Vue.js/PostgreSQL feed parser unifying XML/JSON/CSV/Excel supplier feeds, integrated Facebook Shop and Marketplace APIs, and shipped a CAD/USD FX-rate service tied to inventory dates so cross-border catalogs stayed in sync without double entry
- Technical Skills:** Python, Django, Vue.js, PostgreSQL, Git
- Soft Skills:** Remote collaboration, supplier-facing problem solving

## Wandrian

Full Stack Developer (Co-op → Part-time)

Montréal, QC (On-site)

May 2019 – Jan 2020

- On the Italiarail stack with Railkey Tech, integrated Braintree and extended purchase APIs while rebuilding the Selenium E2E harness and Django/Pyramid/jQuery internal tools for steadier bookings and earlier UI regression catches
- Technical Skills:** Python, Django, Selenium, MongoDB, Git
- Soft Skills:** QA partnership, detail-oriented delivery

## Algolux

Engineering Intern

Montréal, QC (On-site)

May 2017 – Aug 2017; May 2018 – Aug

2018

- Built C++/CUDA/OpenCV imaging tooling for camera/ISP validation (HDR Nexus 6 rigs, Python RAW-color processing, Flask REST capture) and produced an approximately 12,000-image outdoor vision dataset, cutting manual validation time
- Technical Skills:** Python, C++, CUDA, OpenCV, Flask, Linux
- Soft Skills:** Research discipline, clear lab documentation

## Education

### École de technologie supérieure (ÉTS)

Master's Degree, Information Technology Engineering

Montréal, QC, Canada

Aug 2023 – Apr 2026

- Thesis: couples GelSight-style tactile sensing with interactive real-time physics so wear updates friction and shifts texture-space coordinates as the surface geometry changes.

### École de technologie supérieure (ÉTS)

Bachelor's Degree, Information Technology Engineering

Montréal, QC, Canada

Sep 2018 – Apr 2023

- AlgoÉTS (ÉTS algorithmic trading club): President; club infrastructure, web, and trading/data projects.
- Academic focus:** software engineering, distributed systems, databases, and team-based integrator projects in ÉTS labs and coursework.